

Exercise 4: Machine Learning Basics 2

Task 1:

A machine learning model has been trained to classify satellite images into four distinct land-cover types: **Forest (F)**, **Urban (U)**, **Water (W)**, and **Agriculture (A)**.

After running the model on a test set of **100 images**, the following predictions were recorded:

- **Forest (Actual):** 35 total. 30 were correctly predicted as Forest, 2 as Urban, 3 as Agriculture.
- **Urban (Actual):** 20 total. 15 were correctly predicted as Urban, 2 as Forest, 2 as Water, 1 as Agriculture.
- **Water (Actual):** 10 total. 8 were correctly predicted as Water, 2 as Urban.
- **Agriculture (Actual):** 35 total. 25 were correctly predicted as Agriculture, 5 as Forest, 5 as Urban.

1. Construct the Multi-Class Confusion Matrix:

Create a 4 x 4 confusion matrix representing the results above. Rows should represent the Actual Data and columns should represent the Forecast (Predictions).

2. Binary Decomposition:

Break the 4 x 4 matrix down into four separate **2 x 2 Confusion Matrices** (one for each class: F, U, W, and A). In each matrix, the focus class is "1 (Yes)" and all other classes combined are "0 (No)".

3. Calculate Performance Metrics:

Calculate the following metrics for the entire model:

- **Macro-averaged Precision and Recall**
- **Micro-averaged Precision and Recall**
- **Macro-averaged F1-Score**

Implementation Hint:

Coding Option: While you can calculate these values by hand to better understand the underlying logic, you are encouraged to write a small **Python script**. You can use `numpy` for matrix manipulation or the `sklearn.metrics` module (specifically `confusion_matrix` and `classification_report`) to verify your manual results.